Team Tira-miss-u

UNIVERSITY OF PERADENIYA

Department of Computer Engineering, Faculty of Engineering

In collaboration with the Department of Forensic Medicine, Faculty of Medicine

SOFTWARE REQUIREMENTS SPECIFICATION

Forensic Medicine Department Database System

(FMDBS)

Document Type:	Software Requirements Specification (SRS)
Version:	1.0
Date:	June 2026
Client:	Dept. of Forensic Medicine, Faculty of Medicine, UOP
Domain Expert:	Dr. Chathula Wickramasinghe, Act. Consultant JMO

Contents

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Forensic Medicine Department Database System (FMDBS) to be developed by **Team Tira-miss-u** (Department of Computer Engineering, Faculty of Engineering, University of Peradeniya).

The client is the Department of Forensic Medicine, Faculty of Medicine, University of Peradeniya. This document serves as the authoritative reference for all developers, testers, project managers, and stakeholders throughout the software development lifecycle.

1.2 Project Background

The Department of Forensic Medicine at the University of Peradeniya manages two principal categories of forensic work: clinical forensic examinations and autopsy (post-mortem) examinations.

All records, including Medico-Legal Examination Forms (MLEFs), Post-Mortem Reports (PMRs), investigation results, photographs, referral reports, court summons, and certificates of receipt, are currently maintained manually through paper files, handwritten registers, and basic spreadsheets. This paper-based approach creates significant challenges in record retrieval, case tracking, report generation, and inter-departmental communication. The FMDBS aims to digitize and streamline these processes, providing a secure, role-based, searchable digital repository for all departmental records.

1.3 Scope

The FMDBS shall encompass the following high-level functional areas:

- **Clinical Forensic Case Management:** Digital management of all clinical forensic examinations, including MLEFs, patient records, injury documentation, investigation results, referrals, and medico-legal report (MLR) generation.
- **Autopsy Case Management:** Digital management of post-mortem examinations including inquest/court orders, PMRs, cause-of-death forms, photographs, investigation findings, and report dispatch tracking.
- **User Management:** Role-based access control for all departmental users (doctors, clerks, lab staff, photographers, and super administrators).
- **Document Management:** Secure upload, storage, retrieval, and versioning of scanned documents, photographs, X-rays, and laboratory reports.
- **Notification and Communication System:** Automated notifications for pending MLEFs, court dates, overdue reports, and inter-departmental communications.
- **Reporting and Analytics:** Automated report generation using department-approved templates and aggregated analytics for research and audit purposes.

The system will **NOT** manage financial transactions, human resources payroll, or patient clinical treatment records beyond what is documented in forensic examination forms.

1.4 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
FMDBS	Forensic Medicine Department Database System
MLEF	Medico-Legal Examination Form: official form issued by Sri Lanka Police authorising a forensic examination
MLR	Medico-Legal Report: formal report issued by the forensic doctor to the court or police
PMR	Post-Mortem Report: formal report resulting from an autopsy examination
PM	Post-Mortem (autopsy)
JMO	Judicial Medical Officer
BHT	Bed Head Ticket: hospital in-patient record number
ISD	Inquirer into Sudden Deaths: police officer designated to conduct sudden death inquests
COD	Cause of Death
NIC	National Identity Card (Sri Lanka)
RTA	Road Traffic Accident
SRS	Software Requirements Specification
RBAC	Role-Based Access Control
UOP	University of Peradeniya
THP	Teaching Hospital Peradeniya
SLMC	Sri Lanka Medical Council

1.5 References

- Presentation: 'Forensic Medicine Department Database System' by Dr. Chathula Wickramasinghe, Act. Consultant JMO, Dept. Forensic Medicine, UOP (May 2026)
- Sample MLEF forms and body-diagram examination sheets provided by the Dept. of Forensic Medicine
- Sample Medico-Legal Reports (MLRs) and PM Registry photographs provided by the Dept. of Forensic Medicine
- Sample Certificate of Receipt of Medical Report (court acknowledgement documents)
- Requirements elicitation interviews conducted with Dr. Chathula Wickramasinghe, and staff members of the Dept. Forensic Medicine, (May 2026)

2 Overall System Description

2.1 Product Perspective

The FMDBS is a new, purpose-built web-based information system. It does not replace any existing software (none exists) but replaces current manual paper-based processes. The system shall be accessible as a website and via standard desktop application from within the department network and, for authorized users, via a mobile application.

External Interfaces

- Hospital Administration - BHT number reference via manual data entry
- Courts - inquest/court orders and summons received and stored as scanned documents
- Email Services - SMTP gateway for automated notifications

2.2 Product Functions Summary

#	Core Function
1	Secure user authentication and role-based access control (RBAC)
2	Digital case registration for clinical forensic and autopsy cases
3	MLEF and PMR data capture, storage, and retrieval
4	Audio recording and transcript generation
5	Document management (photographs, X-rays, scanned documents, lab reports)
6	Case status tracking (pending reports, pending court dates)
7	Automated medico-legal report generation using templates
8	Notification system for pending actions and court date reminders
9	Research-mode analytics and anonymised data export
10	Extract report case details using the uploaded report correctly in the system
11	Full audit log of all system actions

2.3 User Classes and Characteristics

2.3.1 Super Administrator (Head of Forensic Dept)

Role: Administrative head of the Forensic Medicine Department.
Note: This is a transferable role held by the Head of the Department. The Super Administrator is typically a doctor who can access the system to perform standard doctor duties, but has a separate profile/view for administrative oversight.

- Oversight access to all cases and reports within the department
- User account creation, modification, deactivation, and role assignment
- Access to full audit logs and security reports
- System configuration, maintenance, and database backup/restore

- User management for department-level users (doctors, clerks, photographers, lab staff)
- Access to department-wide analytics and research exports
- Approval workflows for report dispatch; notification configuration
- **Technical:** moderate computer literacy assumed

2.3.2 Doctor (Forensic Medical Officer / JMO)

Role: Forensic doctors conducting clinical examinations and autopsies.

- The only role permitted to open and register **new** cases.
- Access limited to cases assigned to or created by that doctor
- MLEF and PMR data entry, editing, and finalisation
- Create audio recordings and generate transcripts during investigations
- Add reviews to patient case files
- Create referral requests and assign them to the required specialists
- Upload referral reports to patient case files
- Report generation, review, and dispatch
- Access to investigation results and referral reports for own cases
- Receive notifications for own pending cases and upcoming court dates
- **Technical:** basic to moderate computer literacy; system must be highly intuitive

2.3.3 Clerk / Administrative Staff

Role: Departmental administrative and clerical officers.

- Case registration: data entry from **older/historical** paper MLEF forms, MLRs and PMRs into the system. Clerks **cannot** open new, active cases.
- Scanning and uploading documents (MLEF scans, court orders, summons)
- Search for cases on the registry without being able to view the clinical case details
- Recording court summons and certificate of receipt data
- Cannot edit clinical or examination findings
- **Technical:** basic computer literacy; guided UI required

2.3.4 Lab Staff / Technician

Role: Laboratory technicians handling toxicology, histology, and other investigations.

- Upload laboratory investigation results (PDF/image) linked to a case
- Update investigation status (pending / completed)

- No access to clinical examination findings, reports, or case management functions
- **Technical:** basic computer literacy

2.3.5 Dedicated Photographer

Role: Department-assigned photographer for crime scene and post-mortem photography.

- Upload and annotate photographs linked to a specific case
- View photograph gallery for own uploaded cases only
- No access to clinical findings, reports, or other case data
- **Technical:** basic computer literacy; PC upload interface

2.4 Operating Environment

- Desktop (Windows) and mobile applications (Android and iOS) built using Flutter.
- Server hosted on UOP infrastructure or a university-approved cloud platform.
- Responsive design supporting desktop screens (1366×768 and above) and tablet access for the web portal. Mobile access is provided via the dedicated application for authorized users.
- Operating system: Dockerized platform-independent server (Linux recommended).
- Database: relational DBMS (PostgreSQL).
- File storage: secure file server with automated backup (documents, photographs, scanned forms).

2.5 Design and Implementation Constraints

- All patient/victim data is medico-legally sensitive; must comply with Sri Lankan data protection regulations and medical ethics standards.
- The system must support Sinhala language text in free-text fields (Unicode UTF-8).
- Compatible with existing departmental hardware (standard desktop PCs).
- Core functions must be operable on the local hospital network without internet dependency.
- No integration with Sri Lanka Police IT systems in version 1.0; MLEF data entered manually.
- All generated reports must be exportable as PDF.

2.6 Assumptions and Dependencies

- Data entry will be facilitated by scanning documents and generating a preview of required fields through OCR.
- All user roles have access to appropriate devices with working internet connections.
- The dedicated photographer uploads photographs directly using a PC.

- Email notifications require a functioning SMTP server.
- Court summons and inquest orders are received as paper documents and scanned for upload.
- Each hospital has one assigned police station; this mapping is configured by the administrator.

3 Functional Requirements

3.1 User Authentication and Access Control

FR-AUTH-01: User Login

- The system shall provide a secure login interface with username and password.
- The system shall enforce password complexity rules (minimum 8 characters, mixed case, numerals, and special characters).
- The system shall lock an account after 3 consecutive failed login attempts and notify the administrator.
- The system will allow passwords to be reset or reconfigured through a request to the Super Admin.
- The system shall enforce session timeout after a configurable inactivity period for the web application (default: 30 minutes).

FR-AUTH-02: Role-Based Access Control

- Each user shall be assigned exactly one role from the defined set: Super Administrator, Doctor, Clerk, Lab Staff, Photographer. (Note: Super Administrator privileges are assigned to the HoD account).
- Each role shall have a predefined permission set as described in Section 2.3.
- The system shall prevent any user from accessing data or functions not permitted by their role.
- Doctors shall only access cases assigned to or created by them unless they hold an administrator role.

FR-AUTH-03: User Management

- Super Administrator shall create, edit, deactivate, and permanently delete accounts for doctors, clerks, lab staff, and photographers within the department.
- All user management actions shall be logged with a timestamp and performing user identity.

3.2 Clinical Forensic Case Management

FR-CLIN-01: Case Registration

- **Doctors** shall be able to register new, active clinical forensic cases. **Clerks** shall only be permitted to register historical/archived cases from legacy records.
- Registration shall capture:
  - MLEF number
  - Police station
  - Date of MLEF issue
  - Police officer name and registration number

  - Patient full name
  - NIC/passport number
  - Age
  - Gender
  - Address
  - BHT number (if in-patient)
  - Ward (if in-patient)
  - Hospital
  - Admission date/time
  - Reason for examination

- The Reference number for a new case will be calculated, presented in the preview, and subject to modification by the doctor before confirmation.
- The system shall enforce the uniqueness of the Reference number.
- Each case shall be assigned to one doctor at registration; the assigned doctor receives a notification.
- One patient may have multiple forensic cases; the system shall link all cases to that patient's profile.

FR-CLIN-02: MLEF Data Entry (Part B: Medical Findings)

- The system shall record the medical findings of the MLEF as a digital scanned copy.

FR-CLIN-03: Clinical Examination Findings and Investigations

- Doctors shall enter detailed examination findings, including injury description, dimensions, location, and clinical opinion.
- The system shall capture investigation orders (X-ray, CT, blood/urine toxicology, and swabs) linked to the case and track status (ordered/pending/completed).
- The system shall generate referral requests within a given template.
- Investigation results (reports, images) shall be uploadable and stored against the case.
- Referral records shall be stored with the referral date and receiving department, such as:
  - Psychiatry
  - Pediatrics
  - Gynecology/Obstetrics
  - ENT
  - OMF
  - Other
- Inpatient and outpatient review records shall be logged with date, findings, and reviewing doctor.

FR-CLIN-04: Medico-Legal Report Generation

- The system shall auto-generate a draft MLR using a standardized department template, pre-populated with case data.

- Doctors shall review and edit the generated draft before finalization.
- The MLR shall include:
  - Serial number
  - Magistrate/court reference
  - Date of issue
  - Patient identification
  - Place of examination
  - Short history
  - Injuries (nature, cause, and category)
  - Special investigations
  - Opinion
  - Cause of death section (if applicable)
- The finalized MLR shall be exportable as a PDF.
- Dispatch details (date, recipient, method) shall be recorded; the case status shall update to 'Report Dispatched.'

FR-CLIN-05: MLEF Register

- The system shall maintain a digital MLEF register auto-populated from case registrations, including:
  - Reference number
  - Issuing police station
  - Date of issue
  - MLEF number
  - Reason
  - Patient name
  - Age
  - Referral source
  - Referrals made
  - Report issued status
  - Review status
- Department administrators and doctors shall search, filter, and export the MLEF register.
- Clerks may search the register but be unable to view case details.

3.3 Autopsy Case Management

FR-AUTO-01: Autopsy Case Registration

- **Doctors** shall register new autopsy (post-mortem) cases. **Clerks** shall only register historical cases.
- Registration shall capture:

- PM serial number
- Inquest number
- Court
- Case No.
- Date and time of inquest
- Date/time of PM
- Place of PM
- Police station and officer details

- If available, it shall also capture:
  - Deceased's full name
  - NIC/passport
  - Age
  - Gender
  - Address
  - Place of death
  - Place of examination
  - Date/time of death
- If a hospital death:
  - BHT number
  - Ward
  - Preliminary cause-of-death type
- The system shall distinguish hospital deaths from non-hospital outside deaths.

FR-AUTO-02: Inquest Order and Court Order Storage

- The system shall store scanned copies of inquest orders and court orders linked to the autopsy case.
- The following key fields shall be scanned through OCR and be presented as a preview for the Doctor to confirm:
  - Case number
  - Date
  - Deceased details
  - Ordering authority

FR-AUTO-03: Pre-Autopsy Information

- The system shall store pre-autopsy information: crime scene report, BHT and hospital records summaries, statements from family/eyewitnesses, and statements from police.

FR-AUTO-04: Post-Mortem Examination Record (PMR)

- The system shall store the PMR as a scanned copy.
- The system shall provide the ability to audio record the PM and provide an audio transcript of it once completed.

- PM photographs (external, internal) and crime scene photographs shall be uploadable and linked to the case.

FR-AUTO-05: Investigation Management (Autopsy)

- Referrals and Investigations shall be stored as described in FR-CLIN-03.

FR-AUTO-06: Cause of Death Form

- The system shall capture and store the Medicolegal Autopsy Notification Form (COD form) including:
  - Hospital name
  - PM serial number
  - Inquirer details
  - Inquest case number
  - Deceased details
  - Cause of death (immediate, antecedent, contributory)
  - Specimens retained
  - Maternal death status
  - Comments/opinions
  - Autopsy doctor details
- The COD form shall be auto-generated using the Health Ministry template and presented as a preview subject to modification by the Doctor.
- The completed COD form shall be exported as a PDF.

FR-AUTO-07: PM Registry

- The system shall maintain a digital PM registry auto-populated from autopsy case registrations. The registry shall include:
  - Serial number
  - Inquest number
  - Date of inquest
  - PM ordered by
  - Place of PM
  - Police station
  - Government analyst report status
  - Deceased name
  - Cause of death
  - Verdict (accident / suicide / natural / open)

FR-AUTO-08: Post-Mortem Report Generation

- The system shall generate a formal PMR according to a given template pre-populated from entered case data, editable by the doctor before finalization.
- PMR shall be exportable as PDF; dispatch date and recipient shall be recorded.

- Certificate of receipt for PM reports shall be tracked using the same mechanism as FR-CLIN-05.

3.4 Document and Media Management

FR-DOC-01: Document Upload

- Authorized users shall upload documents in JPEG, PNG, and TIFF formats.
- Maximum file size per document: 50 MB (configurable by administrator).
- Documents shall be categorized:
  - MLEF scan
  - PMR scan
  - Photograph
  - X-ray/CT
  - Lab report
  - Toxicology report
  - Histology image
  - Court order
  - Summons
  - Certificate of receipt
  - Other

FR-DOC-02: Document Linking and Retrieval

- All uploaded documents shall be linked to a specific case; multiple documents of the same type may be linked.
- Authorized users shall search and retrieve documents by case number, MLEF/PM number, patient name, date range, and document type.
- Documents shall be viewable in-browser and downloadable.

FR-DOC-03: Document Versioning

- When a document is updated (e.g., amended report), previous versions shall be retained with timestamps and the identity of the uploading user.

3.5 Notification and Communication System

FR-NOTIF-01: In-System Notifications

- The system shall display in-system notifications for:
  - MLEF not yet issued (pending examination)
  - Report not yet dispatched (overdue)
  - Upcoming court dates (7-day, 3-day, and 1-day reminders, configurable)

- ○ Certificate of receipt not received within configured period
- ○ Investigation results uploaded
- ○ New case assigned to doctor

FR-NOTIF-02: SMS Notifications

- Automated SMS/Whatsapp messages shall be sent for the same events as FR-NOTIF-01.
- Message templates shall be configurable by the Department Administrator.
- Doctors shall receive message notifications for their assigned cases only.

3.6 Search and Case Retrieval

FR-SEARCH-01: Case Search

- Authorized users shall search cases by:
  - ○ Case reference number
  - ○ MLEF/PM number
  - ○ Patient name
  - ○ NIC/passport number
  - ○ Police reference number
  - ○ Court reference number
  - ○ Examination date range
  - ○ Case type (clinical/autopsy)
  - ○ Assigned doctor
  - ○ Police station
  - ○ Case status
- Results shall be displayed in a sortable, paginated table.

FR-SEARCH-02: Advanced Filters and Saved Searches

- The system shall support multi-field combined search queries.
- Saved search filters shall be supported for frequently used queries.

3.7 Analytics and Reporting

FR-RPT-01: Statistical Dashboard

- The Super Administrator and doctors shall view a dashboard showing: total cases by type, cases pending reports, cases with pending court dates, cases by reason/category (RTA, assault, sexual assault, etc.), and monthly/annual case volume trends.

FR-RPT-02: Research Access and Export

- Authorized research users shall only access redacted versions of medico-legal documents.

- Personally identifiable information (name, NIC, address, contact details, etc.) shall be automatically concealed before document release.
- Anonymized case data shall be exportable in CSV or Excel format for research.
- Access to original unredacted documents shall require explicit approval from the Super Administrator.

FR-RPT-03: Audit Log

- The system shall maintain a comprehensive audit log of all data access, creation, modification, and deletion events: recording timestamp, user identity, action, and affected record.
- Audit logs shall be accessible only to Super Administrator.

4 Non-Functional Requirements

4.1 Security

NFR-SEC-01 Authentication

- All system access shall require authentication; no anonymous access to case data
- Passwords stored using strong one-way hashing (bcrypt or Argon2)
- 2-factor authentication (2FA) enforced for administrator and high-privilege accounts.

NFR-SEC-02 Authorisation

- Principle of least privilege: users access only data and functions required for their role
- All authorisation decisions enforced server-side; client-side controls are supplementary only

NFR-SEC-03 Data Encryption

- All data in transit encrypted using TLS 1.2 or higher (HTTPS only)
- Sensitive fields (NIC, medical findings) encrypted at rest in the database

NFR-SEC-04 Audit Trail

- Every read, write, update, and delete on case data logged with user identity, timestamp, and IP address
- Audit logs shall be tamper resistant and retained according to institutional and legal retention policies

NFR-SEC-05 Session Security

- Sessions invalidated on logout; session tokens not exposed in URLs
- Concurrent logins from the same account on multiple devices configurable (default: disallowed)
- The system shall enforce session timeout after a configurable inactivity period on the web application (default: 30 minutes).

NFR-SEC-06 Backup and Recovery

- Automated weekly backups shall be maintained securely
- Recovery procedures shall ensure restoration of critical data after system failure or data corruption

4.2 Performance

Requirement	Target
Page load / case search response	≤ 3 seconds under normal load (up to 50 concurrent users)
Report PDF generation	≤ 10 seconds
Document upload	Progress displayed in real-time and does not block other operations
Concurrent users	≥ 100 without performance degradation
Database search (up to 50,000 records)	Results within 3 seconds
System availability (working hours)	Available 24/7 except scheduled maintenance
Planned maintenance notice	≥ 48 hours advance notification to users

4.3 Usability

Requirement	Target
Training time: clerks/photographers	≤ 2 hours to complete primary tasks independently
Training time: doctors (MLEF data entry)	≤ 10 minutes for a standard case after training
Language support	English UI; Sinhala Unicode in free-text fields (UTF-8)
Accessibility standard	WCAG 2.1 Level AA compliance
Screen support	Desktop $\geq 1366 \times 768$, tablet $\geq 1024 \times 768$ (Web Portal). Mobile application available for Doctors.

4.4 Reliability and Data Integrity

- Automated weekly database backups, retained for a minimum of 90 days.
- All data inputs shall be validated; logically inconsistent entries (e.g., examination date before admission date) shall be rejected with a clear error message.
- Graceful degradation: if the notification service is unavailable, core case management functions shall continue unaffected.

4.5 Maintainability

- Built using well-documented, open-source technologies to facilitate long-term maintenance by UOP IT staff.
- Codebase shall include unit tests covering at least 70% of business logic.
- Technical administration manual and user manual delivered with the system.

4.6 Scalability

- Architecture shall support horizontal scaling without significant redesign to accommodate growth.
- Designed to accommodate future integration with government health or justice IT systems through a documented REST API layer.

5 Use Cases

5.1 Actor Summary

Actor	Type	Key Use Cases
Super Administrator	Internal	Manage all users, system config, full audit logs, backup/restore, oversight of all cases, analytics, notification config
Doctor	Internal	Register **new** cases, enter MLEF/PMR data, generate/dispatch reports
Clerk	Internal	Register **historical** cases, data entry, scan/upload documents, manage MLEF register
Lab Staff	Internal	Upload investigation results, update investigation status
Photographer	Internal	Upload and link photographs to cases

5.2 Use Case Descriptions

Use Case ID	UC-01
Name	Register a New Clinical Forensic Case
Actors	Doctor (Primary for new cases), Clerk (Historical cases only)
Precondition	User authenticated with Clerk or Doctor role. Valid MLEF received from police.
Trigger	Patient brought for a forensic examination with an MLEF (or legacy case being digitized).
Main Flow	1. Doctor (or Clerk for legacy) selects 'New Clinical Case'. 2. User selects Outpatient. 3. User Uploads scan of MLEF form. 4. User confirms information extracted from MLEF form. 5. System saves case.
Alternative Flow	Step 3a: MLEF number already exists; system warns user and displays existing case. User confirms or cancels.
Postcondition	New case created; assigned doctor notified; case appears in MLEF register.

Use Case ID	UC-02
Name	Request Referral and Upload Referral report
Actors	Doctor
Precondition	Case registered (UC-01). Doctor assigned to case and authenticated.
Trigger	Doctor determines external specialist input is required during the forensic examination.
Main Flow	1. Doctor opens case record. 2. Doctor goes to the references section. 3. Doctor selects referral type. 4. Doctor selects generate referral. 5. Doctor views preview, modifies data and exports referral letter. Referral status changes to "Pending". 6. Upon receipt of the referral, doctor uploads scanned copy of referral findings. 7. Referral document is stored in the patient case record. 8. Referral status changes to "Completed". 9. Referral document becomes read-only.
Alternative Flow	Step 4a: If referral type is explicitly not mentioned, Doctor selects "Other".
Postcondition	Referral findings saved to case record and status updated.

Use Case ID	UC-03
Name	Generate and Dispatch Medico-Legal Report
Actors	Doctor
Precondition	Examination findings finalised. Investigations complete or accounted for.
Trigger	Doctor ready to issue MLR to court/police.
Main Flow	1. Doctor selects 'Generate Report' from case. 2. System generates draft MLR from template, pre-populated with case data. 3. Doctor reviews and edits the draft. 4. Doctor finalises and approves the report. 5. System generates PDF. 6. Doctor records dispatch details (date, recipient, method). 7. System updates case status; notifies administrator.
Alternative Flow	None
Postcondition	MLR PDF stored in case; dispatch recorded; status updated; notification sent.

Use Case ID	UC-04
Name	Register Autopsy Case and Enter PMR
Actors	Doctor, Lab Staff, Photographer
Precondition	Inquest order or court order received. Doctor authenticated.
Trigger	PM ordered for a deceased person.
Main Flow	1. Doctor selects "New Autopsy Case". 2. Doctor uploads scanned inquest/court order. 3. Doctor confirms preview of autopsy case with deceased details and inquest order data. 4. Doctor enters pre-autopsy information (crime scene, BHT, statements). 5. Doctor performs PM and records voice. 6. Doctor enters examination findings. 7. Doctor generates and dispatches PM report. 8. Photographer uploads PM photographs. 9. Lab staff upload investigation results (toxicology, histology). 10. Doctor generates COD form preview and confirms data.
Alternative Flow	None
Postcondition	Complete autopsy case stored; PM register updated; COD form generated.

Use Case ID	UC-05
Name	Record Court Summons / Certificate of Receipt
Actors	Clerk, Doctor
Precondition	MLR/PMR has been dispatched. Court summons or certificate of receipt received.
Trigger	Physical court document arrives at the department.
Main Flow	1. Clerk or Doctor scans and uploads the document. 2. Links document to relevant case by MLEF/PM number. 3. Enters court date (from summons) or receipt details. 4. System sets notification reminder for court date. 5. Doctor and administrator receive notification.
Alternative Flow	None
Postcondition	Court date recorded; reminder set; case status updated.

6 Constraints and Future Considerations

6.1 Version 1.0 Constraints

- No direct API integration with Sri Lanka Police IT systems.
- No direct integration with hospital HIS for BHT data; manual entry only.

6.2 Future Considerations

- **LLM Integration:** Integration of a local Large Language Model (LLM) to assist in case summarisation and report generation is deferred to a future version.
- **Tamil Language Support:** Support for Tamil language text in free-text fields (Unicode UTF-8) is planned for a future release.

6.3 Data Retention and Legal Compliance

- Medico-legal records are legally significant documents. The system shall retain all case records, reports, and audit logs indefinitely, consistent with Sri Lankan legal and medical records retention requirements.
- Deletion of a case record will not be permitted; records can be hidden but shall require Super Administrator approval and shall be permanently logged in the audit trail.

A Stakeholder Summary

Stakeholder	System Role	Primary Interest / Concern
Dr. Chathula Wickramasinghe	Client / Domain Expert / Super Admin	Efficient report generation, case tracking, notification system
Forensic Doctors (JMOs)	Primary end users	Ease of data entry, accurate report generation
Clerks / Admin Staff	Data entry (legacy cases), document management	Simple intuitive interface, quick case registration
Lab Technicians	Investigation result upload (Blood, Urine)	Easy file upload, clear case reference linkage
Department Photographer	Photograph upload	Quick upload, correct case linkage
Courts (Magistrate/District/High)	External	Timely report dispatch, accurate receipt recording
Psychiatry / SR / MO-Psychiatry	External (No direct system access)	MLEF details and updates on psychological status are handled via the assigning Doctor
UOP, Faculty of Medicine	Institutional oversight	Data security, legal compliance, system reliability
UOP, Dept Of Computer Engineering	Development team	Feasibility, technology stack, project scope

B MLEF Data Fields Reference

The following key data fields are derived from the physical MLEF form used by the Department:

Field Group	Fields Captured
Police Reference	Police station, MLEF number, Date of issue, Police officer name, Rank, Registration No.
Patient Identity	Full name, Address, NIC/Passport number, Age, Sex
Examination Authority	Hospital, Ward, BHT No., Admission date/time, Date/time of examination, Place of examination
Reason for Examination	Free text - reason for referral/examination
Nature of Bodily Harm	Abrasion, Contusion, Laceration, Cut, Fracture, Firearm inj., Burns, Stab, Bite, Dislocation/Subluxation, Explosive inj., Others, None (tick boxes)
Internal Injuries	Free text
Causative Weapon	Blunt, Sharp, Firearm, Explosive devices, Others (tick boxes)
Category of Hurt	Non-grievous / Grievous; Endangers life: Yes/No
Alcohol Examination	Breathing smell, Under influence, Negative
Drug Examination	Consumed, Under influence, Negative
Sexual Assault Examination	Vaginal/hymen penetration signs, Anal penetration signs, Inter-labial penetration signs
Remarks	Free text (Section 22 of MLEF)
Medical Officer	Signature, Full name, qualifications, SLMC Reg No., Designation

C Glossary of Case Types

Case Type	Description
Road Traffic Accident (RTA)	Most common clinical case type. Injuries sustained in motor vehicle collisions
Assault	Injuries from physical attack by another person
Sexual Abuse	Alleged sexual offences requiring forensic examination (Section 18 of MLEF)
Domestic Abuse	Referrals due to suspected intimate partner or family violence
Alcohol Intoxication	Examination for suspected alcohol intoxication
Detainee Examination	Examination of persons in police or judicial custody
Substance Abuse	Examination for suspected drug use
Age Estimation	Forensic determination of a person's age from physical examination
Other	Cases not belonging to the above case types